Inclusions among grade-report families

Arrows mean subset ($\subseteq$).
They point toward the larger set.

All ordered partitions

$\mathcal{A}_n$

$\subseteq$

At most G grades

$\mathcal{A}_{\leq G}$

$\subseteq$

Fixed order ρ, at most G

$\mathcal{A}_\rho \cap \mathcal{A}_{\leq G}$

$\subseteq$

Coarsen score-tie blocks

$\mathcal{C}_{\text{score},\, \leq G}$

$\subseteq$

Exactly G grades

$\mathcal{A}_{=G}$

$1 \leq G \leq n$; ρ refines the score order.
Inclusion need not be strict.
Exact G is a separate restriction.